Airedale Applied Intelligence Challenge · Data Science track

Three build conditions predict which fault a failed unit has

Team AirBenders: Devin, Prateek, Sajjad, Rishik and Arrush

42%

lower log loss than base rates on held-out months (95% CI 39–45%)

33%

lower repair-cost error than the average (95% CI 29–38%)

$100K

to commit now; $625K more after two short pilots

Problem formulation

Five questions: target, unit, method

Question	Target	Unit	Method	Scored by
Which build variables go with each fault?	Fault class	Serial (3,000)	Fisher exact tests, Wilson CIs	p-value, 95% CI
Which fault will a shipped unit have if it fails?	4-class fault	Serial	4 ML models + 3-flag rule vs priors	Log loss, AUC
What will its field repair cost?	Repair $	Serial	Ridge, boosting, two-stage vs mean	MAE, R^2
Which build cells behave alike?	Fault mix	Build cell (72)	k-means, hierarchical	Silhouette, ARI
Which investments, within $750K?	3-yr net savings	Portfolio (178)	Enumeration, Monte Carlo	Net $, P(best)

Not modelled: whether a unit fails. Every serial in the data failed, so there are no negatives. All rates are shares of failures, and FPY or DPMO can't be computed.

Workflow

From four raw files to a decision in six steps

1

Load

Field (D1), config (D3), rework (D4), EOLT (D5)

2

Join

On serial: 3,000 × 1:1. Family, facility and shift agree 100%

3

Reconcile

Map each log to 4 fault classes, 2-of-3 vote

4

Split

Train Jan–Sep 2025, test Oct 2025–Jan 2026

5a

Inference

Fisher tests, log-cost OLS

5b

Classifier

6 models + 3-flag check

5c

Regression

Repair cost, 5 models

5d

Clustering

72 build cells

6

Decide

Score 178 portfolios, Monte Carlo

Steps 5a–5d use only what's known when a unit ships. Python 3 with pandas, scikit-learn and scipy, one script per step; ./run_all.sh rebuilds every number from the raw files in about five minutes.

Data preparation

A 2-of-3 vote reconciles three defect logs

Fault class	Field (D1)	Rework (D4)	EOLT (D5)
Leak	3 leak types	Leak Repair	Refrigerant Leak
Power wiring	Loose Power Wiring	Rewire	Loose Terminal
Controls	Sensor, Comm Fault	Sensor Rewire	Sensor Miswire
Fan rotation	Fan Backwards	Fan Correction	Fan Rotation Error

Label = the class at least two of three logs agree on. Every unit gets one.

91%

of units: all three logs agree. Each log matches the vote 96.7–97.2% of the time.

Where they disagree: 366 night-shift fan corrections in rework that field and EOLT call loose wiring. 98 stay contested.

Sensitivity: refit with rework's labels: log loss 0.707 vs 1.137 baseline, same conclusions.

Where the cost is

Leaks are 26% of failures but 76% of field cost

Share of failures (n = 3,000)



Share of field repair cost ($3.83M)



Fault	Failures	Mean repair	Field cost	Rework cost
Leak	788	$3,670	$2.89M	$0.40M
Power wiring	1,661	$264	$0.44M	$0.84M
Controls	283	$1,387	$0.39M	$0.14M
Fan rotation	268	$384	$0.10M	$0.14M

Statistical inference

Three faults, each under one build condition

Leak

Facility 2 or Supplier B coils

55%

788 of 1,430 failures
vs 0 of 1,570 elsewhere

Fisher p < 1e-140

Controls

Controls revision A

31.7%

283 of 894, CI 28.7–34.8%
vs 0 of 2,106 on Rev B/C

Fisher p = 2e-165

Fan rotation

Night shift

22.4%

268 of 1,199, CI 20.1–24.8%
vs 0 of 1,801 on day shift

Fisher p = 4e-116

Fair comparison: groups differ in size, so we compare rates within each group, with Wilson 95% CIs and Fisher's exact test. Power wiring has no build driver; it appears everywhere. No effect: region, family, disposition, EOLT station.

Association, not proven cause. To confirm before spending: run Rev A beside Rev B on one line; tear down failed Supplier B coils; audit night-shift fan wiring.

Validation design

Train on nine months, test on the next four

Train · Jan–Sep 2025 · n = 2,252. All fitting, plus hyperparameter tuning by 5-fold stratified cross-validation inside this window.

Test · Oct 2025–Jan 2026 · n = 748, kept out of all fitting and tuning.

Features: known at shipment

Product family, facility, coil supplier, controls revision, shift, customer region.

Excluded: would leak the answer

Rework repair type and EOLT failure mode name the same defect as the target. Field cost and downtime exist only after failure.

Why split by time

The model will score future builds. No month-to-month drift (χ^2 p = 0.32). Build date isn't given, so we split on service date.

Fault classifier · test set, n = 748

Three yes/no flags predict as well as any model

Model	Log loss	Skill	Bal. acc.	Acc.	Leak AUC
Class priors (baseline)	1.094	0%	0.25	0.563	0.50
Multinomial logistic	0.694	37%	0.43	0.639	0.86
Logistic + interactions	0.663	39%	0.46	0.650	0.86
Random forest	0.661	40%	0.46	0.662	0.86
Gradient boosting	0.637	42%	0.48	0.664	0.86
Decision tree, depth 4	0.633	42%	0.41	0.659	0.86
Three flags, 8 cells	0.632	42%	0.43	0.666	0.85

The three flags

Facility 2 or Supplier B? Rev A? Night shift? Fit on training months only.

Not borrowed from the test

Each rule holds in the training months alone with zero exceptions.

Uncertainty

Skill 42%, bootstrap 95% CI 39–45%.

Skill = 1 − log loss ÷ baseline. Log loss, because decisions rank units by probability; accuracy rewards always guessing wiring (56%). A pre-set 8-cell model ties the best fitted one, so picking a "best" model on test scores changes nothing.

Repair-cost regression · test set, n = 748

Predicted repair cost: a third less error than the mean

Model	MAE	RMSE	R^2
Global mean (baseline)	$1,310	$1,538	0.00
Mean by family (baseline)	$1,307	$1,523	0.02
Ridge regression	$914	$1,254	0.33
Gradient boosting	$885	$1,224	0.37
Two-stage: P(fault) × cost(fault, family)	$875	$1,230	0.36

What changes cost

1.52×

LegacyX vs other families for the same fault (CI 1.45–1.60). About $318K a year.

Night shift 1.06× (1.02–1.09). Region, facility, supplier, revision: no effect once fault is known.

We use the two-stage model for decisions: lowest MAE (33% below the mean, 95% CI 29–38%), and every prediction splits into which fault and what it costs. Cost effects come from a log-cost OLS on all 3,000 units (R^2 0.90) with bootstrap CIs.

Interpretation · importance and clusters

Four variables drive predictions and segments

Permutation importance: rise in test log loss when shuffled



72 build cells clustered on fault mix, k = 4

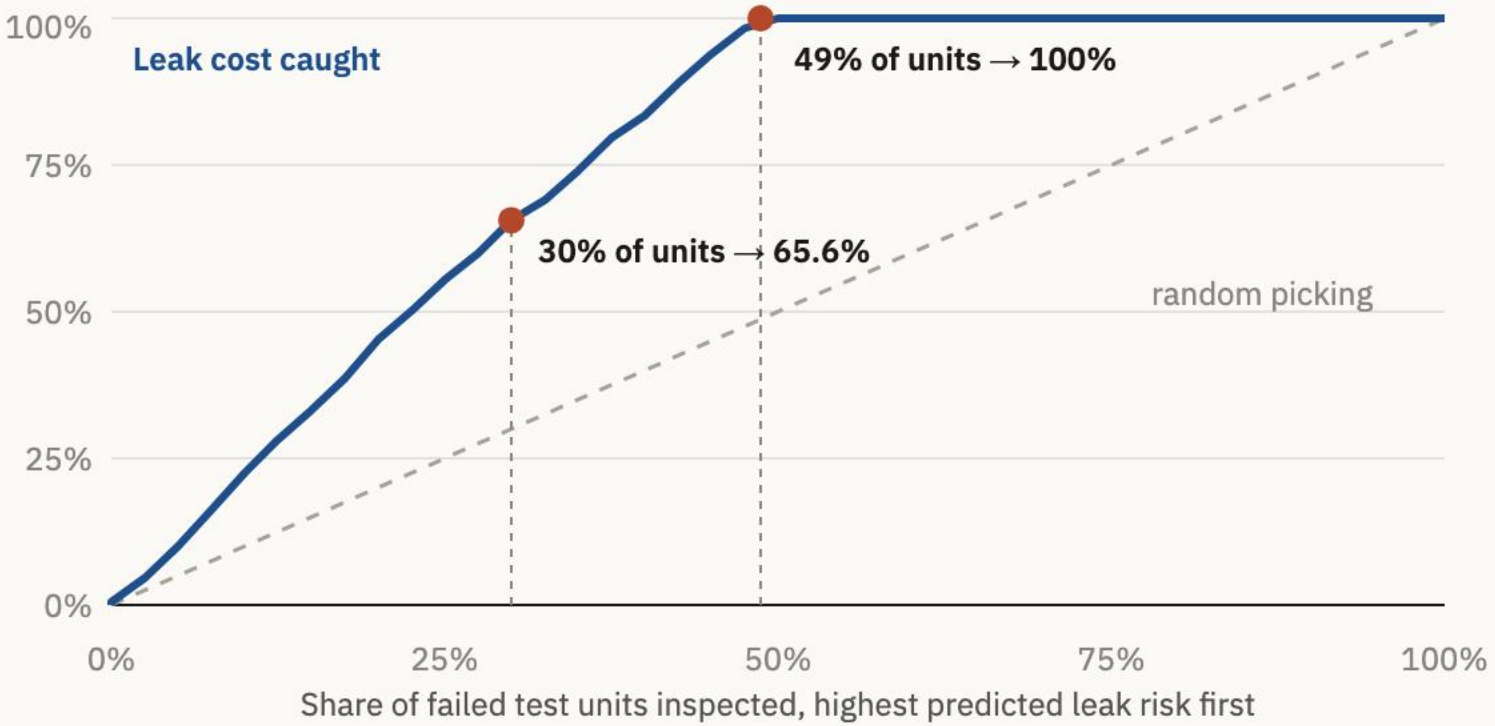
Segment	Units	Main fault	Mean $
Fac 2 or Sup B	1,430	55% leak	$2,208
Rev A, clean source	480	44% controls	$757
Night, Rev B/C	372	38% fan	$320
Everything else	718	98% wiring	$258

Silhouette best at k = 4 (0.69). Hierarchical clustering gives the same split (ARI 1.0).

The clusters weren't told about these variables, yet they recover the same three conditions as the tests. Predictive importance, not causal effect.

Decision impact · which units to flag

Among failures, 30% of units hold 66% of leak cost



Inspecting the 224 riskiest of 748 test units catches **65.6%** of leak cost. Random picking catches 30%.

All leak cost sits in the top 49%: exactly the Facility 2 and Supplier B units.

Decision: scope leak detection to the Facility 2 line and incoming Supplier B coils. We only see failed units, so the rule transfers to the plant; the 30% figure doesn't.

Decision impact · investments within $750K

Best of 178 portfolios: four items, $725K

Investment	Cost	Problem it targets (annual cost today)	Live	Alone, $/yr
Smart Torque Tools	$100K	Power wiring: $0.44M field + $0.84M rework	month 3	$317K
Wiring Poka-Yoke	$125K	Fan + controls miswires: $0.50M + $0.27M	month 4	$308K
Quality Dashboard	$150K	All field escapes: $3.83M	month 6	$574K
Automated Leak Detection	$350K	Leaks: $2.89M field	month 8	$1.16M

$4.60M
3-year net savings

6.3×
3-year ROI

$1.04M
saved in year 1

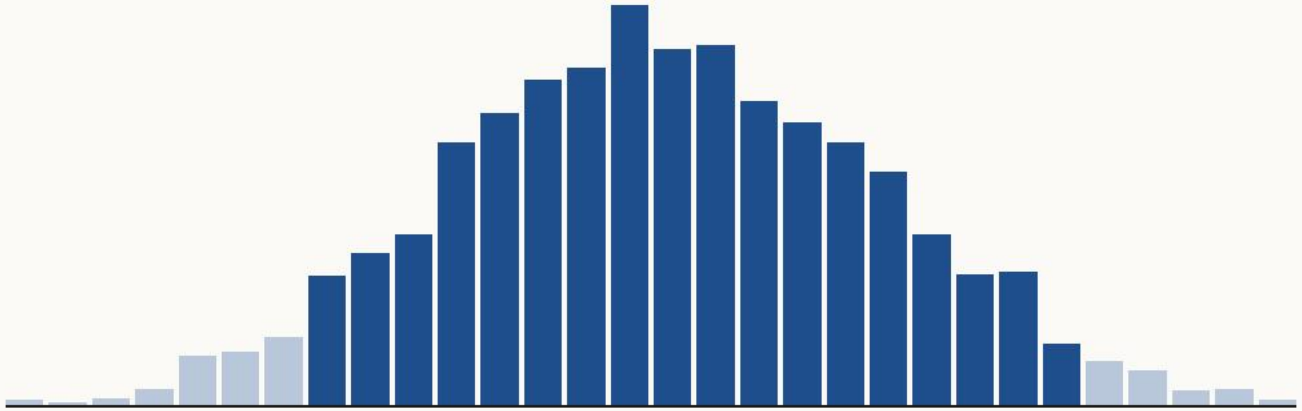
≈1,100
failures a year avoided (est.)

How: each item acts only on the faults and build conditions our tests tie it to, e.g. the supplier program touches only Supplier B leaks. Overlapping benefits multiply rather than add, nothing saves before it's installed, 3-year horizon, no discounting.

Stress test

Holds up under shaky benefits, with one weak spot

3-year net in 2,000 Monte Carlo draws



$3.43M $4.83M

Best of the top 40 in **80.5%** of draws. P5 $3.77M, median $4.14M, P95 $4.53M (dark bars: P5–P95). Never below zero.

Benefit per item varied by risk level (Low ±25%, Medium ±40%, High ±60%) plus 0–3 months of delay. Labor $30–$70/hr: same pick.

The weak point: the Dashboard

We applied its "15% fewer escapes" to all field cost: $574K a year from $150K. At half that benefit it drops out, and the best becomes Torque + Vision Inspection + Leak Detection: $725K, $4.36M net.

All 788 field leaks were caught at EOLT, repaired and passed retest before leaking again. That points at repair quality, not detection.

In every named scenario's winner, and 98% of simulated ones: Smart Torque Tools.

Recommendation

Commit $100K now; release $625K after two pilots

Approve now · $100K

Smart Torque Tools. In the winner of every named scenario and 98% of 2,000 simulated ones; live in 3 months.

Pilot, then fund · $625K

Leak Detection only if a trial catches leaks that EOLT and retest miss; else Braze training, since repairs are brazed, + Supplier program. Then Dashboard + Poka-Yoke or Vision.

Free, outside the catalog

Stop building Rev A (B and C: zero controls faults). Audit night-shift fan wiring. Open a Supplier B corrective action.

Cost of waiting: each month Leak Detection isn't running forgoes about $96K, so the trial should take weeks, not a quarter.

Success measures, monthly: leak share at Fac 2 / Sup B (55%), controls on Rev A (31.7%), fan on nights (22.4%), cost per failure ($1,275).

What our results rely on

Assumption or limit	How we handled it
Every unit failed	No negatives: we model which fault and what cost. Shares are of failures, not of production
No warranty file (D2)	Field repair cost as the warranty proxy; every field event is warranty-eligible
Rework labor rate	Assumed $45/hr; tested $30–$70/hr, same portfolio
Catalog benefits	Used as given; Monte Carlo ±25–60%; Dashboard and detection cut to 50% separately
No build date	Time split on service date; no monthly drift (χ^2 $p = 0.32$)
98 contested labels	Refit with rework's labels: log loss 0.707 vs 1.137, same conclusions
"Best" model picked on test	A pre-set 3-flag model ties it (0.632 vs 0.633), so the pick changes nothing
Observational data	Associations, not proven causes; confirmation steps on slide 6
Tuning folds	Random 5-fold CV inside the training months; safe because there's no drift
Money over time	3-year horizon, undiscounted

Every catalog item, scored alone

Investment	Cost	Months	$/yr saved	3-yr ROI	Payback	In plan
Braze Process Training	$50K	2	$184K	9.5×	5 mo	No*
Smart Torque Tools	$100K	3	$317K	7.7×	7 mo	Yes
Wiring Poka-Yoke	$125K	4	$308K	5.6×	9 mo	Yes
Predictive Quality Dashboard	$150K	6	$574K	8.6×	9 mo	Yes
Enhanced Design Validation	$200K	6	$138K	0.7×	23 mo	No
Vision Inspection	$275K	6	$716K	5.5×	11 mo	Fallback
Automated Leak Detection	$350K	8	$1.16M	6.7×	12 mo	Yes
Additional EOLT Station	$450K	8	$574K	2.0×	17 mo	No
Supplier Quality Program	$175K	4	$369K	4.6×	10 mo	No
LegacyX Product Refresh	$500K	12	$335K	0.3×	30 mo	No

Backup C · leak share of failures by facility and coil supplier

The two leak sources don't stack

	Supplier A	Supplier B	Supplier C
Facility 1	0% · 271	**56% · 221**	0% · 256
Facility 2	**58% · 264**	**57% · 248**	**50% · 272**
Facility 3	0% · 264	**53% · 212**	0% · 276
Facility 4	0% · 240	**57% · 213**	0% · 263

Each cell: leak share of failures · number of failures.

Both sources: 57%. Either alone: 54–55% (p = 0.71).

Fix one source and the 248 units with both stay exposed.

Leak subtype mix is the same for both sources (p = 0.22).